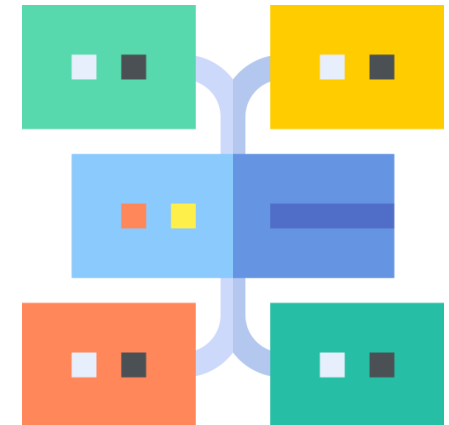




Analisi ed Elaborazione Dati da Sorgenti Multiple

Data integration by a Case Study



Agenda

- Overview of main problems in data integration and management
- Smart Agriculture Case study Part 1: Creation of a uniform dataset
- Smart Agriculture Case study Part 2: Train a machine learning model
- Smart Agriculture Case study Part 3: Ablation Study/**Features Selection**
- Smart Agriculture Case study Part 4: Deploy of an online API

A customer has some data collected in its **smart agriculture** environment.

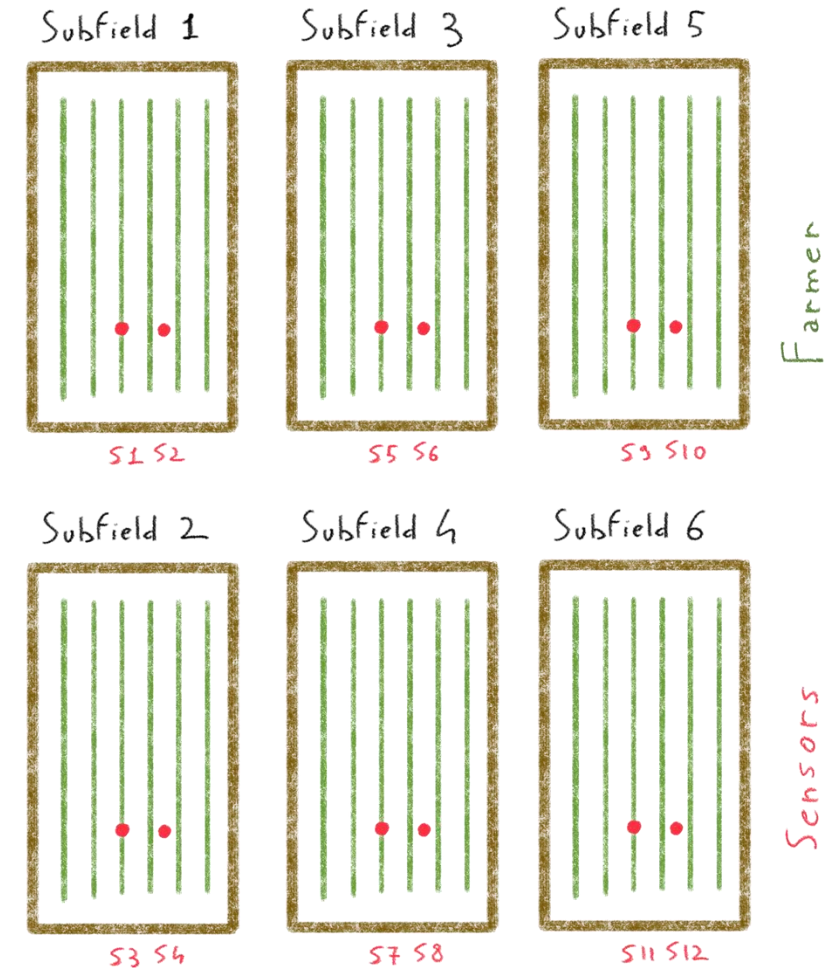
So it want a system capable enable a **best irrigation planning**

- **Goal:** create a service providing reliable soil moisture forecast
- **Steps:**
 - Understand different data sources
 - Provide additional data from reliable sources
 - Perform data harmonization (both for dataset and on-line usage)
 - Investigate different input output data flow
 - Chose a data processing strategy

Case Study: Data Collection Campaign

6 cropping subfield

- **2 Sensors** foreach sub-field
 - 1 Within irrigation line
 - 2 Between irrigation lines
- **2 Irrigation approaches**
 - Farmers experience
 - Sensors threshold



Case Study: available data

The customers provide us with 4 files including data collected during the last cropping season

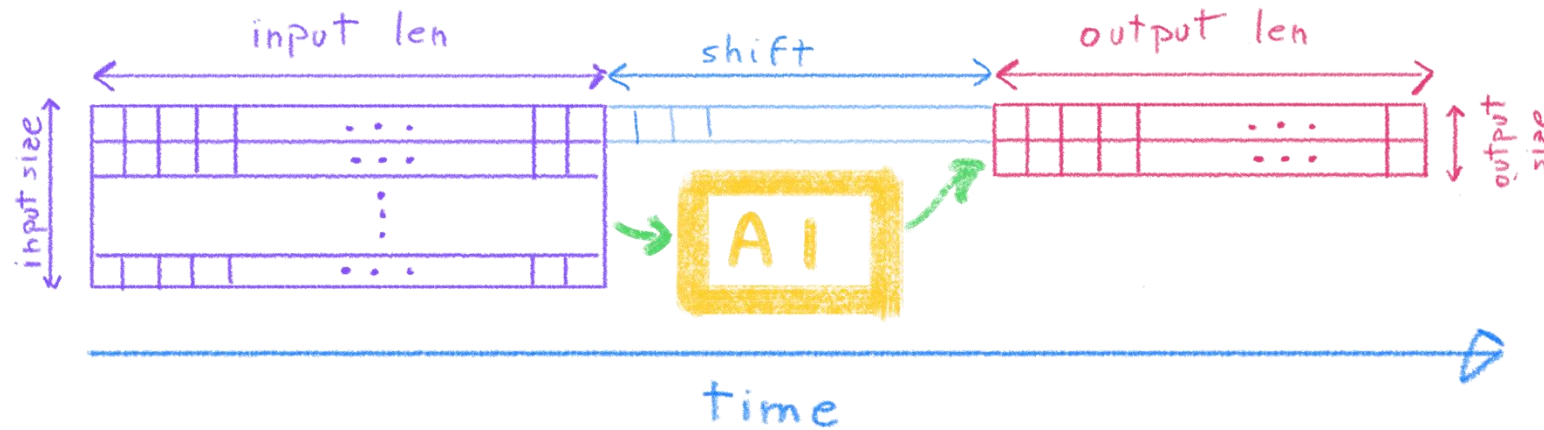
- **Soil_Mositure.xlsx**: file including 12 sheets, each of one contains data for a specific sensor
- **Irrigation_events.xlsx**: file including a list of all the occurred irrigation events, the involved fields and the water volume
- **Meteo_Station.xlsx**: file including weather parameter detected by the on site meteo station
- **LAI.xlsx**: Leaf Area Index sampled for each Irrigation Strategy

Case Study: create Dataset

By the notebook **createDataset.ipynb** in the **CaseStudy** folder of the shared repository we went through several steps devoted to manipulate and reshape the data in order to create 6 csv file, one for each of the subfield.

```
CaseStudy > field_fit
~/My Drive/CNR/Corsi/Adsum/MachineLearning/AdsumMachineLearning/CaseStudy - idity_2m,cloud_cover,wind_speed_10m,wind_direction_100m,soil_temperature_0_to_7cm,soil_temperature_7_to_28cm,soil_temperature_28_to_100cm,rain,
Contains emphasized items
3 2024-12-12 13:00:00,14.1255545008422,13.0105989459292,0.0,0.0001827485380116959,18.9,34,0,7.1,334,15.1,12.2,17.0,0.0,0.0,0.31,16.0,37.0,42135.0,3.0,2.0,316.0,0.0,100590.0
4 2024-12-12 14:00:00,14.1155913637092,12.9260131836708,0.0,0.0003654970760233918,18.9,33,1,7.0,351,16.0,12.5,17.0,0.0,0.0,0.29,19.0,31.0,41185.0,3.0,3.0,324.0,0.0,100440.0
5 2024-12-12 15:00:00,14.2929162209217,12.8714840934774,0.0,0.0005482456140350877,17.8,41,1,9.6,40,16.2,12.8,17.0,0.0,0.0,0.23,19.0,38.0,28431.0,1.0,4.0,15.0,0.0,100380.0
6 2024-12-12 16:00:00,14.2261699736453,12.7692025639253,0.0,0.0007309941520467836,15.9,50,0,8.4,74,15.7,13.1,17.0,0.0,0.0,0.13,18.0,44.0,13915.0,1.0,2.0,54.0,0.0,100400.0
7 2024-12-12 17:00:00,14.1498988876324,13.0062367718832,0.0,0.0009137426900584795,12.4,57,0,6.9,91,14.4,13.3,17.0,0.0,0.0,0.04,18.0,48.0,4766.0,0.0,2.0,63.0,0.0,100410.0
8 2024-12-12 18:00:00,14.3130120346867,13.1245050472597,0.0,0.0010964912280701754,11.4,60,0,3.0,101,12.9,13.4,16.9,0.0,0.0,0.0,14.0,58.0,156.0,0.0,2.0,110.0,0.0,100450.0
9 2024-12-12 19:00:00,14.2171781652268,12.9807655949666,0.0,0.0012792397660818713,10.5,63,0,0.7,78,11.8,13.4,16.9,0.0,0.0,0.0,12.0,60.0,0.0,0.0,1.0,337.0,0.0,100460.0
10 2024-12-12 20:00:00,14.2498257769366,12.960251303345,0.0,0.0014619883040935672,9.6,65,6,1.0,28,10.9,13.3,16.9,0.0,0.0,0.0,10.0,65.0,0.0,0.0,0.0,154.0,0.0,100520.0
11 2024-12-12 21:00:00,14.3463184875287,12.8781193175817,0.0,0.001644736842105263,8.6,67,21,4.1,282,10.2,13.2,16.9,0.0,0.0,0.0,8.0,69.0,0.0,0.0,1.0,180.0,0.0,100570.0
12 2024-12-12 22:00:00,14.2613021590433,12.881054782382,0.0,0.001827485380116959,8.9,64,94,7.3,254,9.9,13.1,16.9,0.0,0.0,0.02,8.0,69.0,0.0,0.0,1.0,177.0,0.0,100620.0
13 2024-12-12 23:00:00,14.2682716667931,13.1807175797357,0.0,0.002010233918128655,10.1,60,72,6.6,242,10.1,13.0,16.9,0.0,0.0,0.02,9.0,68.0,0.0,0.0,1.0,216.0,0.0,100670.0
14 2024-12-13 00:00:00,14.190016225622,12.7537461157662,0.0,0.0021929824561403508,10.4,57,49,7.8,226,10.1,12.9,16.9,0.0,0.0,0.03,6.0,76.0,0.0,0.0,1.0,180.0,0.0,100720.0
15 2024-12-13 01:00:00,14.2462071806647,12.8912927070532,0.0,0.0023757309941520467,9.2,52,26,8.4,222,9.9,13.0,16.9,0.0,0.0,0.03,7.0,76.0,0.0,0.0,0.0,148.0,0.0,100720.0
16 2024-12-13 02:00:00,14.2375160841555,12.9440231967789,0.0,0.0025584795321637425,9.9,54,18,8.5,221,9.9,12.9,16.8,0.0,0.0,0.03,6.0,76.0,0.0,0.0,1.0,180.0,0.0,100730.0
17 2024-12-13 03:00:00,14.0691961879945,13.160843944749,0.0,0.0027412280701754384,8.6,62,23,7.2,224,9.7,12.8,16.8,0.0,0.0,0.02,8.0,67.0,0.0,0.0,0.0,258.0,0.0,100740.0
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20 2024-12-13 06:00:00,14.2579217144178,12.9461028917406,0.0,0.003289473684210526,6.9,76,0,5.5,230,8.9,12.5,16.8,0.0,0.0,0.0,9.0,58.0,0.0,0.0,2.0,192.0,0.0,100780.0
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30 2024-12-13 16:00:00,14.3224588222778,12.9591503377689,0.0,0.005116959064327485,16.0,62,1,12.0,25,16.9,13.8,16.7,0.0,0.0,0.13,19.0,48.0,11847.0,1.0,4.0,36.0,0.0,100990.0
31 2024-12-13 17:00:00,14.2413260985500,12.1163158937460,0.0,0.00530279756230181,12.1,70,2,4.0,40,15.5,14.0,16.6,0.0,0.0,0.02,17.0,50.0,2858.0,0.0,2.0,22.0,0.0,101060.0
```

Case Study: training phase 1



Once the dataset is available we need to define the data flow and structure of input data and forecast (output) data.

- **input len:** Number of past time steps used as input (i.e., the length of the observation window).
- **input size:** number of features in input (soil moisture, irrigation events, solar radiation temperature...)
- **output len:** Number of future time steps to predict (i.e., the length of the forecast horizon)
- **output size:** Number of features to predict in the output sequence (e.g., soil moisture *in* and *between* lines).
- **shift:** Temporal gap (in time steps) between the end of the input window and the start of the prediction window. A shift > 0 introduces a delay between the last observation and the first forecasted point.

Case Study: training phase 1

Network architecture: we have built an **Encoder-Decoder** structure allowing to produce ah-hoc **forecast output** (sequence length, number of features).

The training is provided in the file `runexperiments.py`

